

Autonomous IoT Micro-Controllers for Campus Hostel Print Stations

Transitioning from high-cost, vulnerable laptop deployments to ultra-low-power, tamper-proof **PrintPod** embedded Linux controllers across NERIST campus hostel stations.

Target Institution: North Eastern Regional Institute of Science & Technology (NERIST)	Date: September 2026 · Rev 1.2
Lead Developers: Devananda & PrintKurox Systems Engineering Team	Current Pilot Station: Pare Hostel (Block B, Room 29)

1. Executive Summary & Problem Statement

The **PrintKurox** web platform currently powers autonomous document printing for students at NERIST. At present, the local station spooler operates on a **standard consumer laptop** connected to an Epson EcoTank L3212 printer. While functional for proof-of-concept validation, deploying laptops across multiple hostel blocks incurs severe commercial, physical, and operational vulnerabilities:

Evaluation Metric	Traditional Laptop Deployment	Proposed IoT 'PrintPod' Module	Net Impact
Capital Expenditure (Unit)	■25,000 – ■45,000 / station	■3,850 / station	88% Capex Reduction
Electrical Power Load	45W – 65W continuously (~■250/mo)	3W – 5W continuously (~■20/mo)	92% Energy Savings
Security & Physical Theft Risk	High resale value in student hostel rooms	Enclosed inside tamper-proof chassis bolted to printer	Zero Resale / Theft Target
24/7 Operational Stability	Windows updates, sleep bugs, battery swelling	Industrial Read-Only Linux rootfs with hardware watchdog	99.9% Autonomous Uptime

2. Technical Architecture of the 'PrintPod'

The proposed **PrintPod** hardware module is an embedded ARM Single Board Computer (SBC) designed for unassisted 24/7 operation:

Compute Unit:	Orange Pi Zero 3 (Allwinner H618 Quad-core ARM Cortex-A53 @ 1.5GHz) with 1.5GB LPDDR4 RAM. Effortlessly rasterizes 300/600 DPI PDFs.
Printing Engine:	Native Linux CUPS 2.4+ subsystem paired with the official Epson open-source printer-driver-escpr rasterizer via USB Type-B.
Storage & Integrity:	32GB Class 10 A1 Endurance MicroSD configured with an OverlayFS Read-Only Root Filesystem . Prevents filesystem corruption during sudden hostel power outages.
Student Interface:	1.3-inch I2C Monochrome OLED (128x64) displaying station name, network latency, live 4-digit pickup code, and current print progress, plus dual status LEDs.
Audio Confirmation:	5V active piezo buzzer providing audible chime upon job completion, notifying student in hallway.

3. Campus Network Connectivity & Captive Portal Handling

A critical requirement at NERIST is navigating the **campus Wi-Fi captive web portal** without a mouse or display. The PrintPod solves this via a lightweight, zero-overhead background bot:

1. Autonomous Auto-Login Bot:	Runs an ultra-efficient Python background daemon (0.1% CPU, <12MB RAM). Pings connectivity check endpoints every 30s. If session expires or captive portal redirects, it automatically sends an authenticated HTTP POST with authorized student credentials in 0.2 seconds . <i>Does not overheat or strain the hardware.</i>
2. Institutional MAC Exemption:	Optionally, the board's permanent Wi-Fi MAC address can be registered with the NERIST Computer Center (CC) for institutional IoT device bypass (standard for university printers and lab devices).
3. Redundant 4G LTE Dongle:	For maximum fault-tolerance, a ■1,000 unlocked 4G USB dongle can be attached. PrintKurox only transfers ~50MB daily (under 2GB/month), meaning a basic ■155/mo prepaid recharge provides 100% independent backup when campus Wi-Fi shuts down.

4. Itemized Bill of Materials (BOM) per Station

All components are off-the-shelf and directly orderable from reliable Indian electronics distributors with GST invoicing:

SN	Component & Exact Specification	Trusted Indian Vendor	Qty	Unit Rate	Total (INR)
1	Orange Pi Zero 3 (1.5GB RAM) Allwinner H618 Quad-core, Wi-Fi 5, BT, USB	Robu.in / Silverline Electronics	1	■2,200	■2,200

2	SanDisk 32GB High-Endurance MicroSD Class 10 A1, rated for continuous IoT write cycles	Amazon India / Local Distributor	1	■420	■420
3	5V 3A USB-C Regulated Power Supply BIS-certified, short-circuit & thermal surge protected	Robu.in / MakerBazar	1	■350	■350
4	1.3" I2C OLED Display Module 128x64 White, SH1106 controller, 4-pin I2C	Robu.in / MakerBazar	1	■260	■260
5	AV Feedback (Piezo Buzzer + Dual LEDs) 5V active buzzer, 5mm Green/Blue LED, DuPont wire	Local Electronics Market	1 set	■80	■80
6	USB-A to USB-B Printer Data Cable (0.5m) Shielded short cable for clean enclosed mounting	Amazon India	1	■140	■140
7	Custom Laser-Cut Acrylic / 3D Enclosure Bolts to printer chassis; security screws	NERIST 3D Lab / Local Fabrication	1	■300	■300
8	Hardware Kit & Heat Sinks Self-adhesive copper heat sinks, standoffs, zip ties	Robu.in	1 set	■100	■100
CORE BASE STATION TOTAL (Wi-Fi Enabled)		Plug & Play Kiosk	1	—	■3,850
9	[Optional] 4G LTE USB Dongle (Unlocked) For independent cellular backup (Jio/Airtel)	Amazon India / Flipkart	1	■1,100	+■1,100
10	[Optional] 5V/12V Mini DC UPS Battery Backup Maintains print jobs through hostel load-shedding	Amazon India (Resonate / Oakter)	1	■850	+■850
FULLY LOADED UNIT (Wi-Fi + 4G LTE + Battery Backup)		Autonomous Mission-Critical	1	—	■5,800

5. Phased Campus Rollout Budget Scenarios

Deployment Phase	Scope / Target Locations	Units	Base Budget (Wi-Fi)	Loaded Budget (+4G)	Est. Traditional Laptop Cost
Phase 1: Pilot Validation	Pare Hostel (Block B, Room 29)	1	■3,850	■4,950	■35,000 (1 Laptop)
Phase 2: Core Hostel Ring	Hostel Blocks A, B, C, and D	4	■15,400	■19,800	■1,40,000 (4 Laptops)
Phase 3: Full Campus Grid	All Hostels (A–H, Girls Hostel, Central)	10	■38,500	■49,500	■3,50,000 (10 Laptops)

■ **Key Return on Investment (ROI):** Deploying 4 PrintPod units across Hostel Blocks A, B, C & D costs only **■15,400**, yielding an immediate capital expenditure savings of **■1,24,600** compared to laptops. Furthermore, with each unit drawing only ~4W, annual electrical savings total **~■3,100 per station**, allowing each hardware unit to fully pay for itself within 14 months purely on power efficiency.

6. Implementation Schedule (4-Week Delivery Roadmap)

Timeline	Milestone & Key Deliverables	Expected Completion
Week 1	Procurement of Orange Pi Zero 3, MicroSDs, displays, and power adapters via Robu.in / Amazon.	Day 1 – Day 7
Week 2	OS flashing (DietPi / Debian Lite), CUPS printing subsystem setup, and Epson ESC/P-R driver configuration.	Day 8 – Day 14
Week 3	Integration of PrintKurox Python daemon, I2C OLED display driver, audio chime, and campus auto-login script.	Day 15 – Day 21
Week 4	Enclosure assembly, final 48-hour continuous stress printing in Pare Hostel (Block B), and handover to custodian.	Day 22 – Day 28

Prepared By:	Recommended By (Hostel Custodian):	Approved By (Admin / Incubation Cell):
Devananda & Engineering Team Lead Project Developer, PrintKurox	Hostel Warden / Custodian Pare Hostel (Block B), NERIST	Authorized Signatory Dean of Student Affairs / TBI NERIST